

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Paper No. STB-201: Descriptive Statistics and Distribution Theory

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

BS/Sem 11/35 ST Paper : STB-201 Regular & X - Student (Descriptive Statistics and Distribution Theory) receipt of this question paper.] Note: Answer any five questions. Each question carries equal marks. **Question 1.**

- (a) Discuss the situations under which rank correlation is used. Establish the formula for Spearman's rank correlation when ranks are untied. Obtain the limits of the rank correlation coefficient.
- (b) Prove that the coefficient of correlation is zero when two random variables are independent, and show also by giving an example that the converse is not true. Let r be the coefficient of correlation between two random variables X and Y . Show that:

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

where σ_x^2 , σ_y^2 , and σ_{x-y}^2 are the variances of the random variables X , Y , and $X - Y$ respectively.

Question 2.

- (a) Differentiate between correlation and regression. If $(X_i, Y_i), i = 1, 2, \dots, n$ are pairs of observations, then find the constants a and b of the line $Y = a + bX$ by the least squares method.
- (b) Let Y_e be the predicted value of Y from the regression line of Y on X and $e_i = Y_i - Y_{ei}$ be the error. Show that the variance of error is $V(e) = \sigma_y^2(1 - r^2)$, where σ_y^2 is the variance of Y and r is the coefficient of correlation between X and Y .
- (c) Let $2Y - X + 3 = 0$ be the regression line of Y on X and its angle with the regression line of X on Y is 45° . Obtain the regression coefficient corresponding to the line of X on Y and the correlation coefficient r .

Question 3.

- (a) Define partial correlation. In case of three variables X_1, X_2, X_3 , show that:

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

- (b) Let the random variable X have the marginal density:

$$f(x) = \frac{1}{2}, \quad -1 < x < 1$$

and 0 otherwise. Let the conditional density of Y given $X = x$ be:

$$f(y|x) = \begin{cases} 1 & \text{if } x < y < x+1, \quad -1 < x < 0 \\ 1 & \text{if } -x < y < 1-x, \quad 0 \leq x < 1 \end{cases}$$

Show that the variables X and Y are uncorrelated.

Question 4.

- (a) Define Poisson distribution. Derive the Poisson distribution as a limiting case of the binomial distribution.
- (b) Let $X \sim P(\lambda_1)$ and $Y \sim P(\lambda_2)$ be two independent Poisson random variables. Find the probability distribution of $X + Y$, and find $P(X + Y > 0)$.
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Question 5.

- (a) Differentiate between binomial and hypergeometric distributions.
- (b) A bag contains 3 defective and 4 good items. 3 items are chosen randomly from the bag one by one. If X denotes the number of defective items drawn, write the probability distribution of X under the situations of:
- (i) with replacement,
 - (ii) without replacement.

Show that the probability of selection of a defective item remains the same at each draw in both cases of selection.

Question 6.

- (a) What is meant by attribute? Define the term class, class frequency, and order of class. Given that $(AB) = 150$, $(A\beta) = 230$, $(\alpha B) = 260$, and $(\alpha) = 2340$, find the rest of the frequencies and check the independence of attributes A and B .
- (b) Let X be a binomial variate with parameters n and p . Obtain the moment generating function of X , and show that $E(X/n) = p$ and $Var(X/n) = \frac{p(1-p)}{n}$.
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Question 7.

- (a) Define negative binomial distribution. Obtain the mean and variance of the negative binomial distribution.
 - (b) Define uniform distribution in the continuous case. Obtain its mean and variance. The occurrence time of buses at a stop has a uniform distribution with mean arrival time 2 minutes and variance $4/3$ minutes. What is the probability that a person will wait at least 3 minutes for a bus?
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Question 8.

- (a) Write the probability density function of the normal distribution. Let $X \sim N(\mu, \sigma^2)$. Obtain the moment generating function of X .
- (b) State and prove Chebyshev's inequality. If for a random variable X , $P(|X - 2| < 4) \geq \frac{21}{25}$, then find the mean and variance of X using Chebyshev's inequality.
- (c) Define Beta distribution and find its mean and variance.
- (d) Let X and Y be two independent random variables from:
 - (i) Binomial distribution, i.e., $X \sim B(n_1, p)$ and $Y \sim B(n_2, p)$,
 - (ii) Poisson distribution, i.e., $X \sim P(\lambda_1)$ and $Y \sim P(\lambda_2)$.

Recognize the conditional distribution of $(X = x \mid X + Y = t)$ in each case.